Mutual Fund Analytics Capstone Project

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1. Executive Summary

The Mutual Fund Analytics Capstone Project was developed to perform comprehensive analysis of mutual fund industry data and generate actionable insights related to fund performance, investor behavior, market trends, and portfolio risk. The project aimed to simulate a real-world financial analytics workflow by integrating data engineering, exploratory data analysis, risk assessment, and business intelligence techniques.

The project utilized multiple datasets containing mutual fund information, Net Asset Value (NAV) history, Assets Under Management (AUM), Systematic Investment Plan (SIP) inflows, investor transactions, portfolio holdings, and benchmark indices. These datasets were processed and analyzed using Python, Pandas, NumPy, SQLite, and Power BI. Data cleaning and transformation techniques were applied to ensure data quality, consistency, and reliability before performing analytical tasks.

Exploratory Data Analysis (EDA) was conducted to identify trends in NAV movements, AUM growth, SIP inflows, investor demographics, geographic investment distribution, and category-wise fund inflows. Multiple visualizations were created to uncover patterns and support data-driven decision making. The analysis highlighted significant growth in SIP participation, increasing investor engagement, and varying performance characteristics across mutual fund categories.

Fund Performance Analytics was performed by calculating key financial metrics including daily returns, Compound Annual Growth Rate (CAGR), Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown. A composite fund scorecard model was developed to rank mutual funds based on risk-adjusted performance and investment efficiency. Benchmark comparisons were also conducted to evaluate fund performance relative to market indices.

Advanced analytics techniques were implemented to assess portfolio risk and investor behavior. Value at Risk (VaR) and Conditional Value at Risk (CVaR) were calculated to estimate potential downside risk. Rolling Sharpe Ratio analysis provided insights into changing risk-adjusted returns over time. Investor cohort analysis and SIP continuity analysis were used to understand investment patterns and identify investors at risk of discontinuing systematic investments. Additionally, a simple fund recommendation system was developed to suggest suitable mutual funds based on investor risk profiles.

To support business users and decision makers, an interactive Power BI dashboard was created consisting of four pages: Industry Overview, Fund Performance, Investor Analytics, and SIP & Market Trends. The dashboard enables dynamic exploration of key metrics, trends, and performance indicators through interactive visualizations and filters.

Overall, the project demonstrates the application of data analytics, financial analysis, and business intelligence techniques in the mutual fund domain. The resulting insights can assist investors, fund managers, and financial institutions in evaluating performance, managing risk, and making informed investment decisions.

2. Project Objectives

The primary objective of this project was to analyze mutual fund industry data and derive meaningful insights that support investment decision-making, risk assessment, and business intelligence reporting. The project was designed to simulate a real-world financial analytics workflow by combining data engineering, exploratory data analysis, performance evaluation, risk modeling, and dashboard development.

The specific objectives of the project were:

1. To collect, clean, validate, and organize mutual fund datasets from multiple sources for analytical processing.
2. To design and implement a structured SQLite database for efficient storage and retrieval of mutual fund data.
3. To perform Exploratory Data Analysis (EDA) on NAV history, Assets Under Management (AUM), SIP inflows, investor demographics, portfolio holdings, and category-wise fund inflows.
4. To evaluate mutual fund performance using key financial metrics such as Daily Returns, CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown.
5. To develop a fund ranking and scoring framework that compares mutual funds based on risk-adjusted performance indicators.
6. To analyze investor behavior through cohort analysis, SIP continuation tracking, and transaction pattern analysis.
7. To assess portfolio and market risk using Value at Risk (VaR), Conditional Value at Risk (CVaR), Rolling Sharpe Ratios, and Sector Concentration Risk (HHI).
8. To build a simple recommendation engine capable of suggesting suitable mutual funds based on investor risk preferences.
9. To create interactive Power BI dashboards that provide a comprehensive view of industry trends, fund performance, investor analytics, and SIP market dynamics.
10. To present findings, insights, and recommendations through a professional report and business intelligence dashboard that supports data-driven investment decisions.

By achieving these objectives, the project demonstrates the practical application of data analytics, financial modeling, and visualization techniques within the mutual fund investment domain.

3. Data Sources and Dataset Overview

The Mutual Fund Analytics project utilized ten datasets representing different aspects of the Indian mutual fund industry. These datasets were used to perform data cleaning, exploratory analysis, performance evaluation, risk assessment, investor behavior analysis, and dashboard development.

The data was structured to simulate a real-world mutual fund analytics environment and included information related to fund schemes, NAV history, investor transactions, SIP inflows, portfolio holdings, benchmark indices, and industry trends.

Table 3.1: Dataset Overview

Dataset Name	Description	Key Columns
Fund Master	Master information about mutual fund schemes and fund houses	amfi_code, scheme_name, fund_house, category, plan
NAV History	Historical daily NAV values for all schemes	amfi_code, date, nav
AUM by Fund House	Assets Under Management by fund house over time	date, fund_house, aum_crore
Monthly SIP Inflows	Monthly SIP contributions and growth statistics	month, sip_inflow_crore, active_sip_accounts_crore
Category Inflows	Net inflows across mutual fund categories	month, category, net_inflow_crore
Industry Folio Count	Growth in investor folios across fund categories	month, total_folios_crore
Scheme Performance	Fund return and risk performance metrics	return_1yr_pct, return_3yr_pct, sharpe_ratio, alpha
Investor Transactions	Individual investor transaction records	investor_id, transaction_type, amount_inr, state
Portfolio Holdings	Sector and stock allocation of fund portfolios	stock_name, sector, weight_pct
Benchmark Indices	Historical benchmark index values	date, index_name, close_value

4. ETL and Data Cleaning Process

The ETL (Extract, Transform, Load) process was implemented to ensure that all datasets used in the Mutual Fund Analytics project were accurate, consistent, and suitable for analysis. Data cleaning and validation were performed using Python, Pandas, and SQLite before conducting any analytical tasks.

4.1 Data Extraction

The project utilized ten datasets containing mutual fund information, NAV history, AUM statistics, SIP inflows, investor transactions, portfolio holdings, benchmark indices, and industry metrics. These datasets were imported into Python using the Pandas library and stored in CSV format.

The extracted datasets included:

- Fund Master Data
- NAV History Data
- AUM by Fund House
- Monthly SIP Inflows
- Category Inflows
- Industry Folio Counts
- Scheme Performance Metrics
- Investor Transactions
- Portfolio Holdings
- Benchmark Index Data

4.2 Data Validation

Data validation was performed to identify inconsistencies and ensure data quality. The following checks were conducted:

- Verification of mandatory fields.
- Validation of numerical columns such as NAV values, returns, AUM, and transaction amounts.
- Verification of valid AMFI codes across datasets.
- Validation of transaction types and investor attributes.
- Verification of benchmark and portfolio allocation values.

Any records failing validation rules were reviewed and corrected where applicable.

4.3 Handling Missing Values

Missing values were identified using Pandas functions such as `isnull()` and `info()`.

The following techniques were applied:

- Forward filling missing NAV values to account for market holidays and non-trading days.
- Replacing missing categorical values where appropriate.
- Removing records with critical missing information.
- Ensuring no null values remained in key analytical columns.

This process improved data completeness and analytical reliability.

4.4 Duplicate Removal

Duplicate records were identified using the `duplicated()` function.

Examples included:

- Duplicate NAV entries for the same fund and date.
- Repeated investor transaction records.
- Duplicate benchmark observations.

Duplicate records were removed to prevent inaccurate calculations and reporting.

4.5 Date Conversion and Standardization

Several datasets contained date fields in different formats. These were standardized using Pandas datetime functions.

Examples include:

- NAV dates
- Transaction dates
- SIP reporting periods
- Benchmark index dates

Date standardization enabled accurate time-series analysis, trend identification, and performance calculations.

4.6 Data Transformation

Additional transformations were performed to support analysis:

- Calculation of daily returns from NAV values.
- Creation of cohort year variables for investor analysis.
- Generation of rolling performance metrics.
- Conversion of percentage values into analytical measures.
- Aggregation of transaction and inflow data for reporting purposes.

These transformations provided structured datasets for advanced analytics and dashboard development.

4.7 SQLite Database Design

A relational SQLite database was created to store cleaned and processed data.

The database schema consisted of multiple tables, including:

- dim_fund
- fact_nav
- fact_transactions
- fact_performance

Primary and foreign key relationships were established using the AMFI code as the common identifier across datasets.

The database enabled efficient querying, storage, and retrieval of mutual fund information for analytical processing.

4.8 ETL Outcome

The ETL process successfully transformed raw datasets into clean, structured, and analysis-ready data assets. The cleaned datasets were stored in CSV format and loaded into SQLite for further analysis. This ensured consistency across all subsequent stages of the project, including Exploratory Data Analysis (EDA), performance analytics, risk assessment, and dashboard development.

5. Exploratory Data Analysis (EDA)

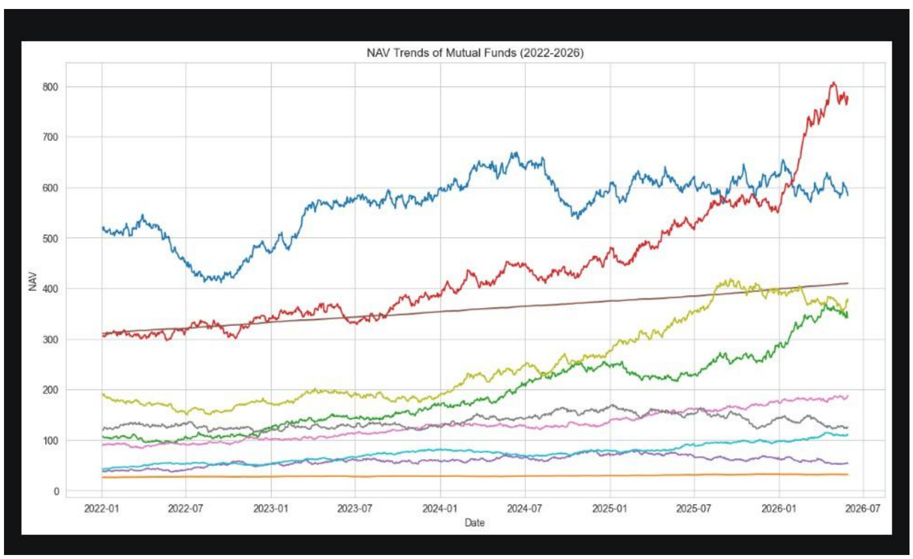
Exploratory Data Analysis (EDA) was conducted to understand the characteristics of the mutual fund industry, identify trends, detect patterns, and generate meaningful insights from the available datasets. Various statistical techniques and visualizations were used to analyze NAV movements, Assets Under Management (AUM), SIP inflows, investor demographics, geographic distribution, and portfolio allocation.

5.1 NAV Trend Analysis

Historical Net Asset Value (NAV) data was analyzed to study the performance of mutual fund schemes over time. Daily NAV values from 2022 to 2025 were plotted to observe growth trends, market fluctuations, and recovery periods.

Observations

- Most equity-oriented funds exhibited long-term growth trends.
- Market corrections resulted in temporary declines in NAV values.
- Recovery periods demonstrated resilience across major fund categories.
- Large-cap funds generally displayed lower volatility compared to small-cap funds.



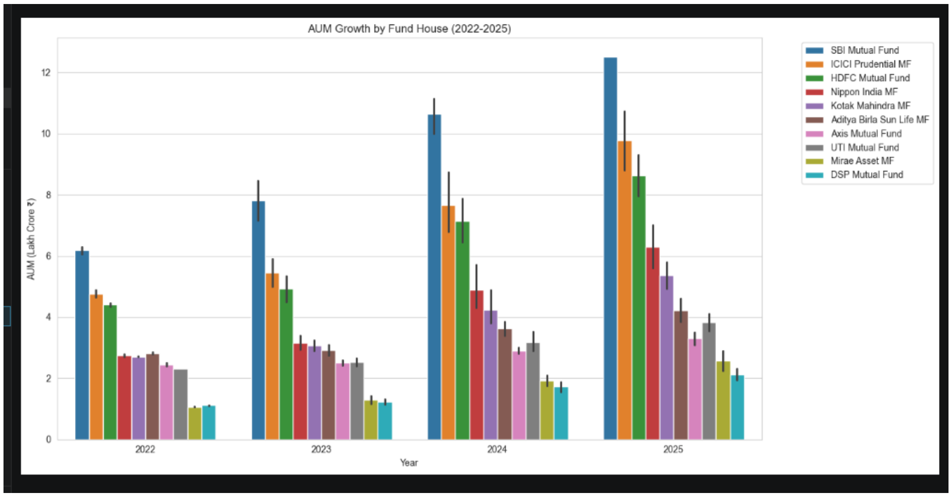
5.2 Assets Under Management (AUM) Growth Analysis

AUM data was analyzed to evaluate the growth of mutual fund companies over the study period. Fund houses were compared based on total assets managed.

Observations

- SBI Mutual Fund maintained the highest AUM among all fund houses.

- Significant growth was observed across major asset management companies.
- Increasing investor participation contributed to rising AUM levels.
- Large fund houses continued to dominate the market.

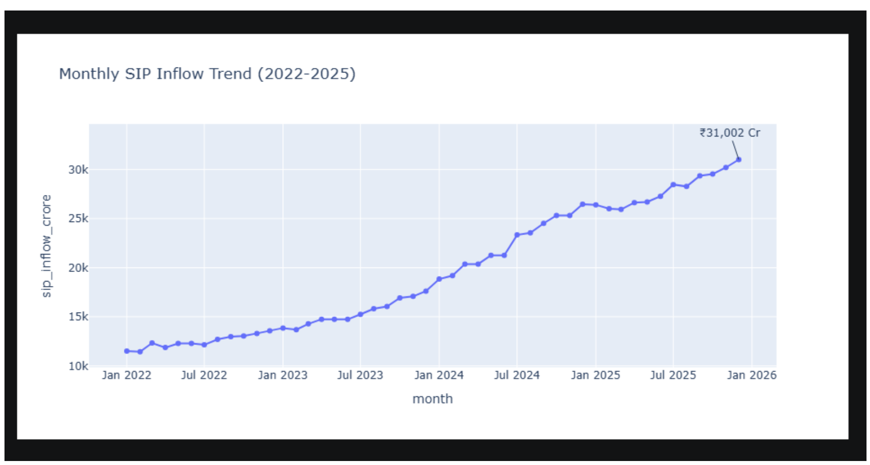


5.3 SIP Inflow Analysis

Monthly SIP inflows were analyzed to understand investment trends and investor confidence in systematic investing.

Observations

- SIP inflows showed consistent growth during the analysis period.
- Investor preference for disciplined investing increased significantly.
- Active SIP accounts grew steadily over time.
- Monthly SIP contributions reached record levels by the end of the study period.



5.4 Category-wise Inflow Analysis

Category inflows were examined to identify investor preferences across mutual fund categories.

Observations

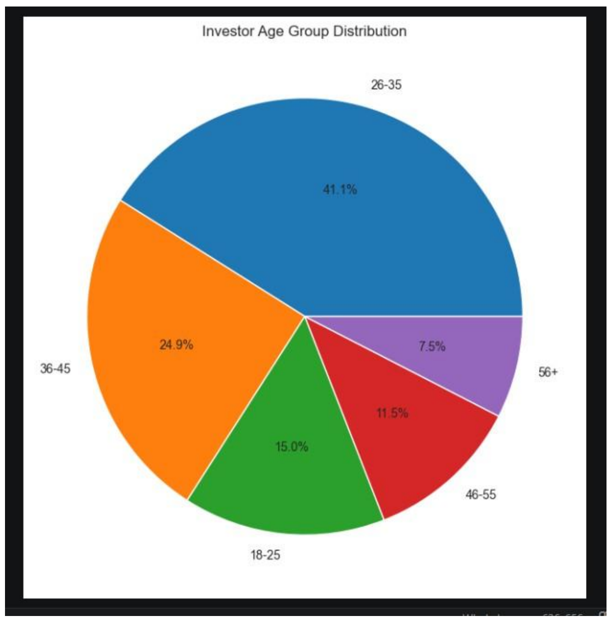
- Flexi Cap and Mid Cap funds attracted significant inflows.
- Large Cap funds continued to receive stable investments.
- Certain categories experienced temporary fluctuations due to market conditions.
- Investors increasingly diversified investments across multiple categories.

5.5 Investor Demographic Analysis

Investor transaction data was analyzed to study demographic participation patterns.

Observations

- The 26–35 age group represented the largest investor segment.
- Younger investors showed greater adoption of SIP investments.
- Investment participation declined gradually in higher age groups.
- Long-term wealth creation emerged as a key investment objective.



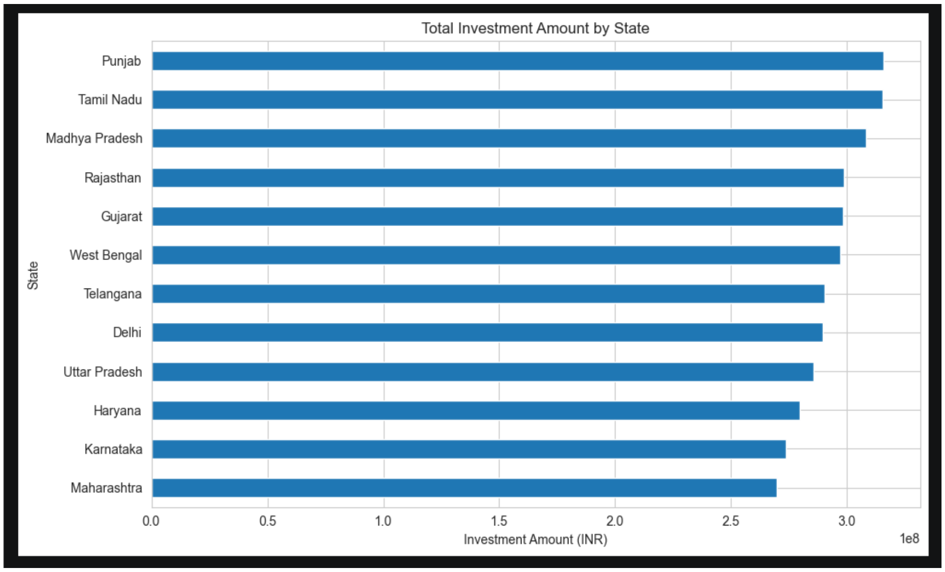
5.6 Geographic Distribution Analysis

State-wise investment activity was analyzed to identify regional participation trends.

Observations

- Investments were concentrated in major urban and economically developed states.

- Punjab, Madhya Pradesh, Tamil Nadu, Gujarat, and Karnataka showed strong participation.
- Both T30 and B30 cities contributed significantly to mutual fund investments.
- Geographic diversification of investors improved over time.

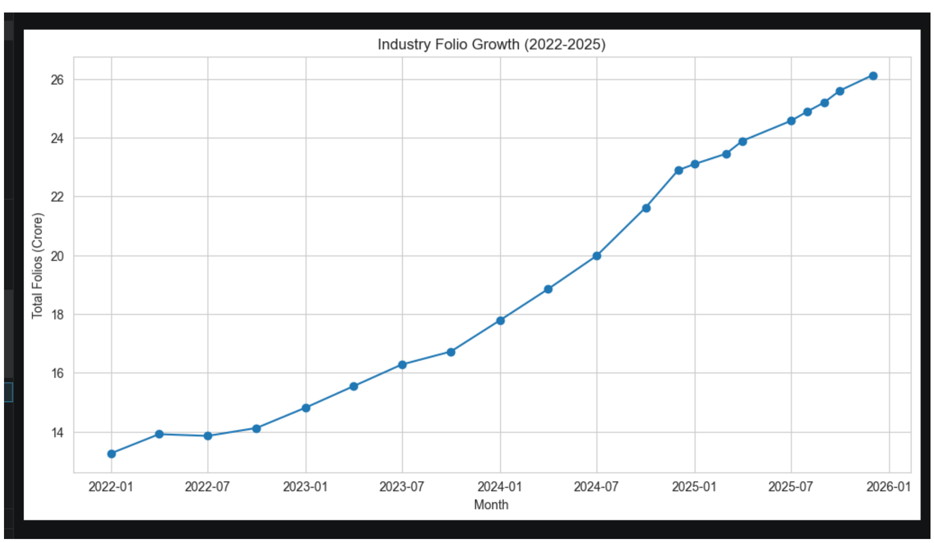


5.7 Folio Growth Analysis

Industry folio counts were analyzed to measure investor participation and market penetration.

Observations

- Total folio counts increased consistently throughout the study period.
- Equity folios represented the largest share of total folios.
- Growth in folio counts indicated increasing retail investor participation.
- The mutual fund industry experienced strong expansion.

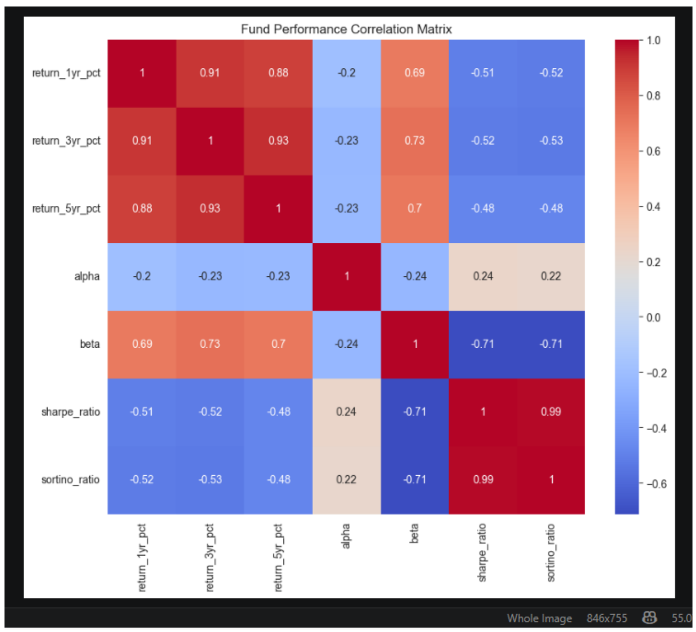


5.8 Correlation Analysis

Correlation analysis was performed on selected mutual fund returns to understand relationships between fund performances.

Observations

- Funds within similar categories exhibited positive correlations.
- Diversification opportunities existed across different fund categories.
- Market-wide movements influenced multiple funds simultaneously.
- Correlation analysis supported portfolio diversification strategies.

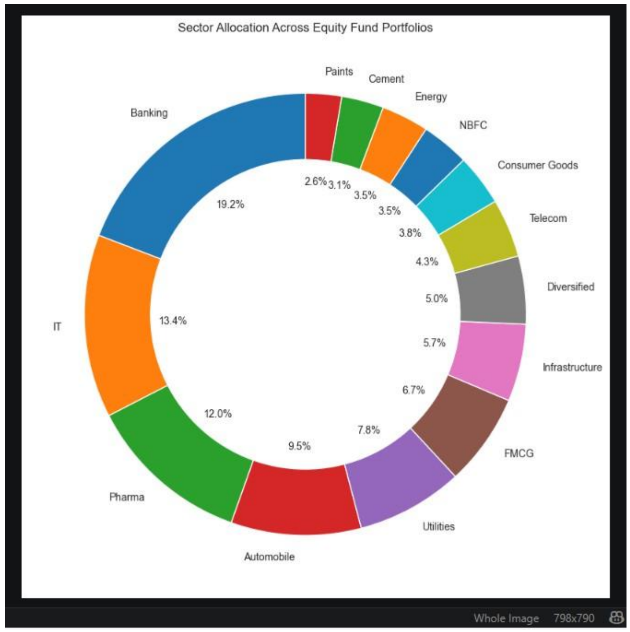


5.9 Sector Allocation Analysis

Portfolio holdings data was used to analyze sector-wise investment allocations.

Observations

- Banking and Financial Services accounted for a significant portion of holdings.
- Technology and Healthcare sectors maintained strong representation.
- Sector diversification varied across funds.
- Portfolio composition reflected prevailing market trends.



5.10 Key EDA Findings

The exploratory analysis generated several important insights:

1. SIP inflows demonstrated sustained growth, indicating increasing investor confidence.
2. SBI Mutual Fund maintained the highest Assets Under Management.
3. The 26–35 age group was the most active investor segment.
4. Equity-oriented funds dominated investor preferences.
5. Flexi Cap and Mid Cap categories attracted strong inflows.
6. Folio counts increased steadily, reflecting industry growth.
7. Geographic participation expanded across multiple states.
8. Mutual fund returns exhibited varying levels of correlation.
9. Portfolio diversification differed significantly across funds.
10. Overall industry indicators highlighted a positive growth trajectory for the mutual fund sector.

6. Fund Performance Analytics

Fund Performance Analytics was conducted to evaluate the effectiveness, consistency, and risk-adjusted returns of mutual fund schemes. Various financial performance indicators were calculated using historical NAV data and benchmark information. These metrics provide investors with a comprehensive understanding of fund performance beyond simple return comparisons.

6.1 Daily Returns Analysis

Daily returns were calculated using historical NAV values to measure day-to-day changes in fund performance.

Formula

Daily Return = (NAVt / NAVt-1) - 1

Daily return analysis serves as the foundation for calculating advanced performance and risk metrics such as Sharpe Ratio, Sortino Ratio, VaR, and Maximum Drawdown.

Observations

- Daily returns varied across different fund categories.
- Equity-oriented funds exhibited higher volatility compared to debt-oriented funds.
- Market corrections resulted in temporary negative return periods.
- Long-term trends remained positive for most funds.

6.2 Compound Annual Growth Rate (CAGR)

CAGR was calculated to evaluate the annualized growth rate of investments over multiple years.

Formula

CAGR = (Ending NAV / Beginning NAV)^(1/n) - 1

Where:

- n = Number of years

Observations

- Several large-cap funds delivered stable CAGR values.
- Long-term investment horizons generally resulted in better annualized returns.
- Growth-oriented funds demonstrated higher CAGR compared to conservative funds.

- CAGR provided a standardized measure for comparing funds with different investment periods.

  amfi_code,cagr,scheme_name

  100016,0.026370743672111008,HDFC Top 100 Fund - Regular Plan - Growth

  100025,0.04458209619173137,HDFC Short Term Debt Fund - Regular - Growth

  100033,0.3012315310072722,HDFC Mid-Cap Opportunities Fund - Regular - Growth

  101206,0.2353836133468803,ABSL Frontline Equity Fund - Regular - Growth

  101207,0.07938764837247714,ABSL Small Cap Fund - Regular - Growth

  101208,0.06509026855465327,ABSL Liquid Fund - Regular - Growth

  102885,0.18232378887705947,UTI Nifty 50 Index Fund - Regular - Growth

  102886,0.011717454918341197,UTI Mid Cap Fund - Regular - Growth

  102887,0.16841120247307528,UTI Flexi Cap Fund - Regular - Growth

6.3 Sharpe Ratio Analysis

The Sharpe Ratio was used to evaluate risk-adjusted returns by comparing excess returns against overall portfolio volatility.

Formula

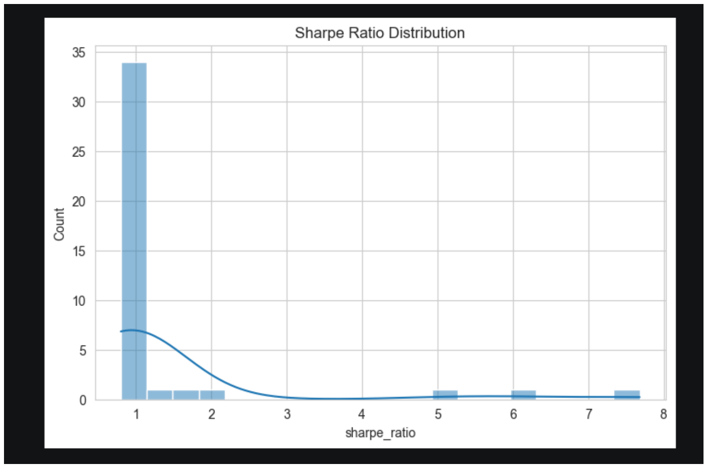
Sharpe Ratio = (Rp - Rf) / σp

Where:

- Rp = Portfolio Return
- Rf = Risk-Free Rate
- σp = Portfolio Standard Deviation

Observations

- Funds with higher Sharpe Ratios generated superior returns relative to risk.
- Moderate-risk funds demonstrated strong risk-adjusted performance.
- Sharpe Ratio enabled fair comparison between funds with different volatility levels.
- HDFC Top 100 Fund and Mirae Asset Large Cap Fund ranked among the best-performing moderate-risk funds.



6.4 Sortino Ratio Analysis

The Sortino Ratio was calculated to focus specifically on downside risk rather than total volatility.

Formula

Sortino Ratio = (Rp - Rf) / Downside Deviation

Observations

- Sortino Ratios provided a more investor-focused measure of risk.
- Funds with limited downside volatility achieved higher Sortino values.
- Several funds demonstrated strong downside risk management.
- Sortino analysis complemented Sharpe Ratio findings.

amfi_code,sortino_ratio

100016,-0.47376091238348456

100025,-0.9361664680297559

100033,1.980757029261388

101206,1.939282932029156

101207,0.07582654302337254

101208,-0.9202353836970717

102885,1.5112493982837234

102886,-0.4994892859579716

102887,1.0823889558046962

6.5 Alpha and Beta Analysis

Alpha and Beta were calculated relative to benchmark indices to measure performance and market sensitivity.

Alpha

Alpha represents excess returns generated beyond benchmark expectations.

Beta

Beta measures sensitivity of a fund to market movements.

Observations

- Positive Alpha values indicated fund managers generated value beyond market returns.
- Beta values close to 1 suggested market-like behavior.
- Higher Beta funds experienced greater market sensitivity.
- Alpha and Beta helped evaluate active fund management effectiveness.

amfi_code,alpha,beta

100016,0.03747580763443281,-0.0582684314388089

100025,0.042817920315060876,0.0011582183155281463

100033,0.2719535450128533,0.0051035258305283104

101206,0.21399784776051212,0.02108631625012803

101207,0.10897091901704538,-0.06528924788351524

101208,0.060861457957871144,0.00026670968338238433

6.6 Maximum Drawdown Analysis

Maximum Drawdown measures the largest decline from a fund's peak value during the analysis period.

Formula

Maximum Drawdown = Minimum(NAV / Running Maximum NAV - 1)

Observations

- Funds with lower drawdowns demonstrated stronger downside protection.
- Drawdown analysis highlighted vulnerability during market corrections.
- Conservative funds generally experienced lower drawdowns.

- Maximum Drawdown served as an important risk indicator.

```
amfi_code,max_drawdown
100016,-0.24734440880673303
100025,-0.04308263824636349
100033,-0.16217208574644848
101206,-0.11291596329323073
101207,-0.3544691649336603
101208,-0.0016224980244717857
102885,-0.1085986215250232
102886,-0.2800112356639287
102887,-0.21539827112383225
```

6.7 Fund Scorecard Model

A composite scoring model was developed to rank mutual funds using multiple performance indicators.

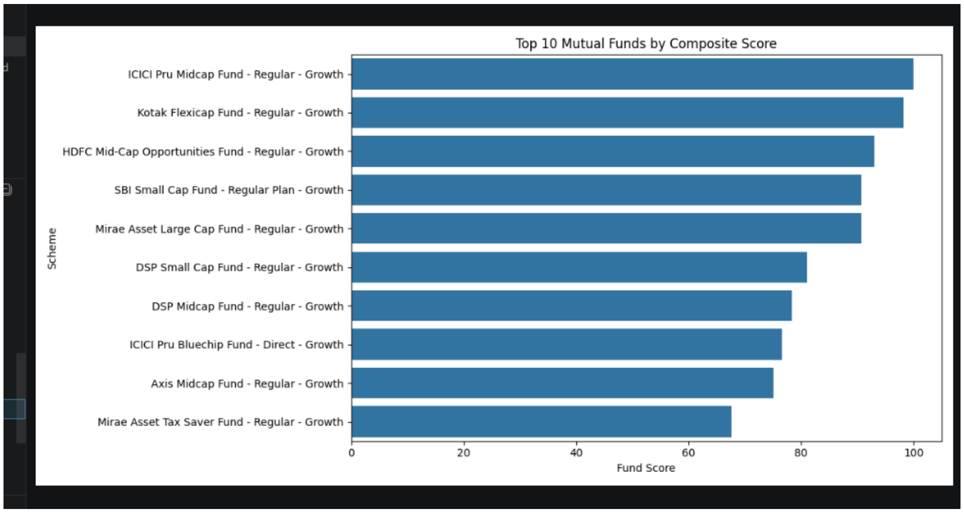
Score Components

- 30% Weight: 3-Year Return Rank
- 25% Weight: Sharpe Ratio Rank
- 20% Weight: Alpha Rank
- 15% Weight: Expense Ratio Rank (Inverse)
- 10% Weight: Maximum Drawdown Rank (Inverse)

The scorecard produced a standardized score between 0 and 100 for each fund.

Benefits

- Simplifies fund comparison.
- Combines performance and risk metrics.
- Provides objective ranking criteria.
- Supports investment decision-making.

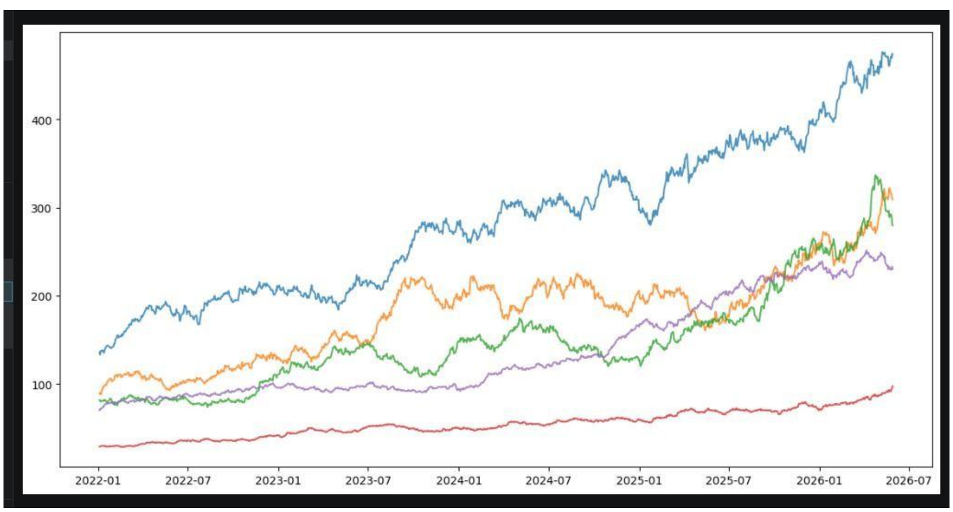


6.8 Benchmark Comparison

Top-performing funds were compared against benchmark indices such as NIFTY 50 and NIFTY 100.

Observations

- Several funds consistently outperformed benchmark returns.
- Benchmark comparisons highlighted active management effectiveness.
- Tracking error analysis measured deviation from benchmark performance.
- Long-term performance remained competitive relative to market indices.



6.9 Key Findings

1. Risk-adjusted metrics provided deeper insights than return analysis alone.
2. Moderate-risk funds achieved strong Sharpe and Sortino Ratios.
3. Several funds generated positive Alpha, indicating superior management performance.

4. Maximum Drawdown analysis highlighted differences in downside protection.
5. Composite fund scoring enabled objective ranking of investment opportunities.
6. Benchmark comparisons demonstrated the ability of select funds to outperform market indices.
7. Long-term performance metrics supported informed investment decision-making.

7. Advanced Analytics and Risk Metrics

Advanced analytics techniques were implemented to evaluate portfolio risk, investor behavior, and fund suitability. These analyses extend beyond traditional performance metrics and provide deeper insights into potential investment risks, investor engagement patterns, and portfolio diversification.

7.1 Value at Risk (VaR)

Value at Risk (VaR) was calculated to estimate the maximum expected loss that a mutual fund may experience under normal market conditions at a specified confidence level.

For this project, Historical VaR was calculated using the 95% confidence level.

Formula

VaR (95%) = 5th Percentile of Daily Return Distribution

Purpose

- Estimate downside risk.
- Quantify potential losses during adverse market conditions.
- Compare risk exposure across different funds.

Observations

- Funds with higher volatility exhibited larger VaR values.
- Equity-oriented funds generally showed greater downside risk than conservative funds.
- VaR provided a useful measure for comparing risk profiles across schemes.

7.2 Conditional Value at Risk (CVaR)

Conditional Value at Risk (CVaR) was calculated to estimate the average loss beyond the VaR threshold.

Formula

CVaR = Mean(Returns Below VaR Threshold)

Purpose

- Measure extreme downside risk.
- Assess potential losses during severe market conditions.
- Complement VaR analysis by evaluating tail-risk exposure.

Observations

- CVaR values were consistently lower than VaR values.
- Funds with large downside volatility exhibited higher tail risk.
- CVaR provided additional insight into worst-case scenarios.

amfi_code,VaR_95,CVaR_95

100016,-0.014363639988111142,-0.01806020763217176

100025,-0.003793246199522682,-0.004993763020560281

100033,-0.019033535861558825,-0.023455762770703794

101206,-0.013281661986281489,-0.017439425687129195

101207,-0.026021248221061066,-0.03245905898704435

101208,-0.00026865751304001\u0035\u0039,-0.00042208206558576\u0038\u0035

102885,-0.01261262882069356,-0.01549040525311211

102886,-0.01922028344426503,-0.02325086026922\u0031\u0032\u0031\u0033

102887,-0.015231617279157872,-0.019411182679158694

7.3 Rolling Sharpe Ratio Analysis

A 90-day Rolling Sharpe Ratio was calculated for selected funds to evaluate changes in risk-adjusted performance over time.

Formula

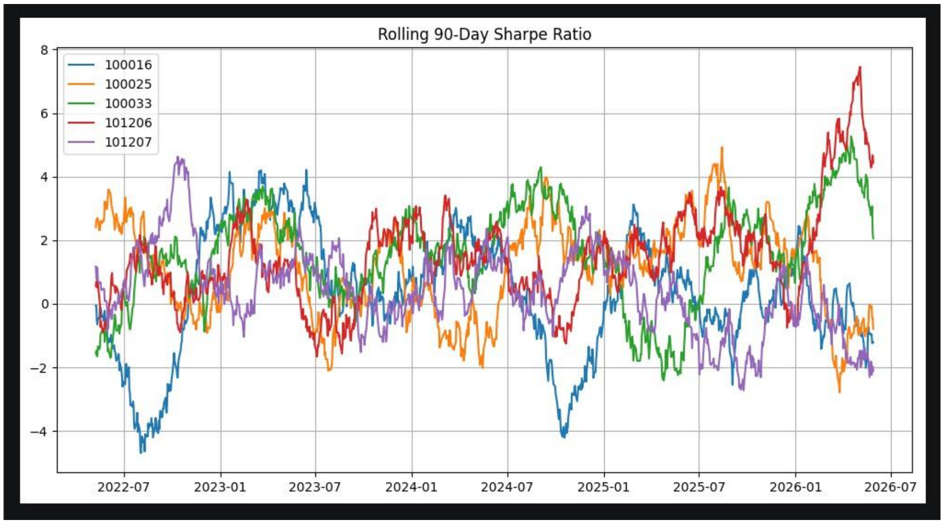
Rolling Sharpe Ratio = (Rolling Mean Return / Rolling Standard Deviation) × √252

Purpose

- Monitor performance consistency.
- Evaluate changing market conditions.
- Compare risk-adjusted returns across different periods.

Observations

- Sharpe Ratios fluctuated based on market performance.
- Strong market rallies improved risk-adjusted returns.
- Periods of market uncertainty resulted in lower Sharpe values.



7.4 Investor Cohort Analysis

Investor Cohort Analysis was conducted by grouping investors according to the year of their first investment transaction.

Objectives

- Understand investor retention patterns.
- Compare investment behavior across cohorts.
- Measure average investment contributions.

Observations

- Recent investor cohorts contributed significant investment volumes.
- Average investment amounts varied across cohorts.
- Cohort analysis revealed differences in investor participation and engagement.

ohort_year,avg_investment,total_investment

2024,107422.54183205638,3491125187

2025,109158.5770609319,30455243

7.5 SIP Continuation Analysis

SIP Continuation Analysis evaluated the consistency of investor contributions through Systematic Investment Plans (SIPs).

Investors with more than six SIP transactions were analyzed to calculate the average gap between contributions.

Criteria

- Average Gap ≤ 35 Days → Active Investor
- Average Gap > 35 Days → At-Risk Investor

Observations

- Most investors maintained regular SIP contributions.
- A subset of investors displayed irregular investment patterns.
- Early identification of at-risk investors can support retention strategies.

```
investor_id,avg_gap,sip_count,status

INV000001,76.0,2,At Risk

INV000002,207.0,3,At Risk

INV000003,238.0,2,At Risk

INV000004,85.4,6,At Risk

INV000005,14.0,3,Active

INV000006,123.75,5,At Risk

INV000007,104.5,5,At Risk

INV000008,70.4,6,At Risk

INV000009,85.33333333333333,4,At Risk
```

7.6 Fund Recommendation System

A simple recommendation engine was developed to suggest mutual funds based on investor risk preferences.

Inputs

- Low Risk
- Moderate Risk
- High Risk

Recommendation Logic

Funds were filtered based on risk grade and ranked according to Sharpe Ratio.

Sample Recommendation Output

Fund Name	Risk Grade	Sharpe Ratio
HDFC Top 100 Fund	Moderate	1.06
Mirae Asset Large Cap Fund	Moderate	1.06
ICICI Pru Bluechip Fund	Moderate	1.03

Benefits

- Supports personalized investment selection.
- Simplifies fund discovery.
- Utilizes risk-adjusted performance measures.

7.7 Sector Concentration Risk Analysis (HHI)

Sector concentration risk was evaluated using the Herfindahl-Hirschman Index (HHI).

Formula

$HHI = \Sigma(Weight_i^2)$

Where:

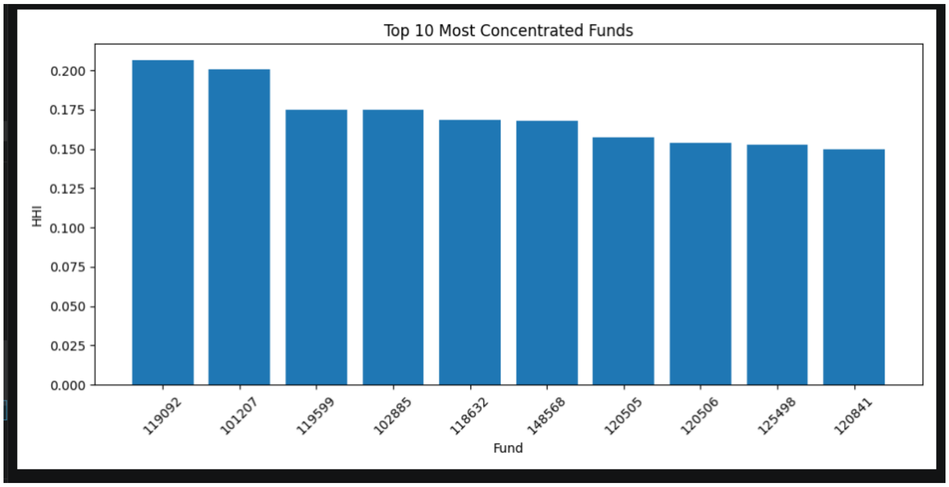
- $Weight_i$ represents sector allocation weight.

Interpretation

- Higher HHI → More Concentrated Portfolio
- Lower HHI → Better Diversification

Observations

- Some funds exhibited significant concentration in Banking and Financial Services sectors.
- Diversified portfolios showed lower HHI values.
- Sector concentration can increase portfolio-specific risk.



7.8 Key Findings

1. VaR and CVaR analysis identified funds with greater downside risk exposure.
2. Rolling Sharpe Ratios highlighted periods of strong and weak risk-adjusted performance.
3. Investor cohorts demonstrated varying investment behaviors and contribution levels.
4. SIP Continuity Analysis successfully identified potentially inactive investors.
5. The recommendation system generated suitable fund suggestions based on risk profiles.
6. Sector concentration analysis revealed differences in portfolio diversification levels.
7. Advanced analytics provided valuable insights beyond traditional return-based evaluation.

8. Dashboard Development and Visualization

An interactive Power BI dashboard was developed to transform analytical findings into visually intuitive and business-friendly insights. The dashboard integrates multiple datasets and analytical outputs to provide a comprehensive view of mutual fund industry performance, investor behavior, risk metrics, and market trends.

The dashboard was designed with interactivity, filtering capabilities, and performance monitoring features to support decision-making by investors, analysts, and fund managers.

8.1 Dashboard Objectives

The primary objectives of the dashboard were:

- Provide a consolidated view of mutual fund industry performance.
- Monitor Assets Under Management (AUM) and SIP growth.
- Compare fund performance across different categories.
- Analyze investor demographics and transaction behavior.
- Visualize market trends and investment flows.
- Enable interactive exploration through slicers and filters.

The dashboard was built using Microsoft Power BI and consisted of four dedicated analytical pages.

8.2 Dashboard Page 1: Industry Overview

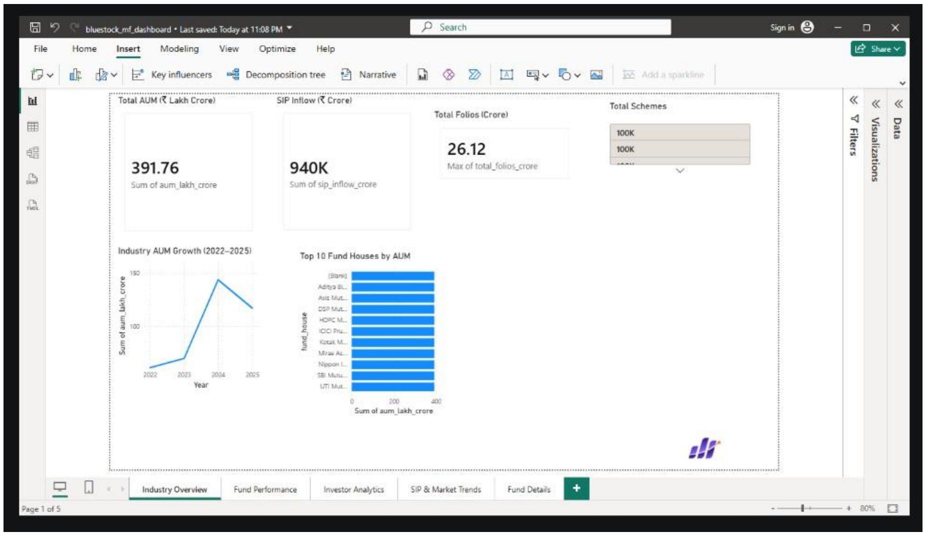
The Industry Overview page presents high-level metrics related to the mutual fund industry.

Key Components

- Total Assets Under Management (AUM)
- Total SIP Inflows
- Total Investor Folios
- Number of Mutual Fund Schemes
- Industry AUM Growth Trend
- Top Fund Houses by AUM

Insights

- AUM demonstrated significant growth throughout the study period.
- SBI Mutual Fund maintained the highest market share.
- SIP participation increased consistently over time.
- The mutual fund industry experienced strong expansion.



8.3 Dashboard Page 2: Fund Performance Analytics

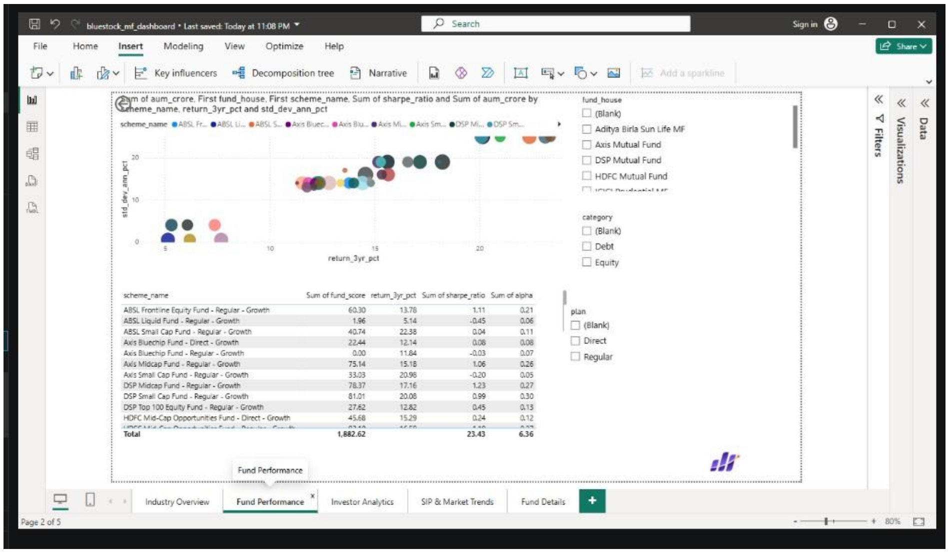
This page focuses on evaluating mutual fund performance using risk-adjusted metrics and comparative analysis.

Key Components

- Return vs Risk Scatter Plot
- Fund Scorecard Table
- Fund Category Filters
- Fund House Filters
- Investment Plan Filters

Insights

- High-performing funds achieved strong risk-adjusted returns.
- Moderate-risk funds demonstrated favorable Sharpe Ratios.
- Risk-return relationships were clearly visible through scatter analysis.
- Interactive filtering enabled detailed fund comparison.



8.4 Dashboard Page 3: Investor Analytics

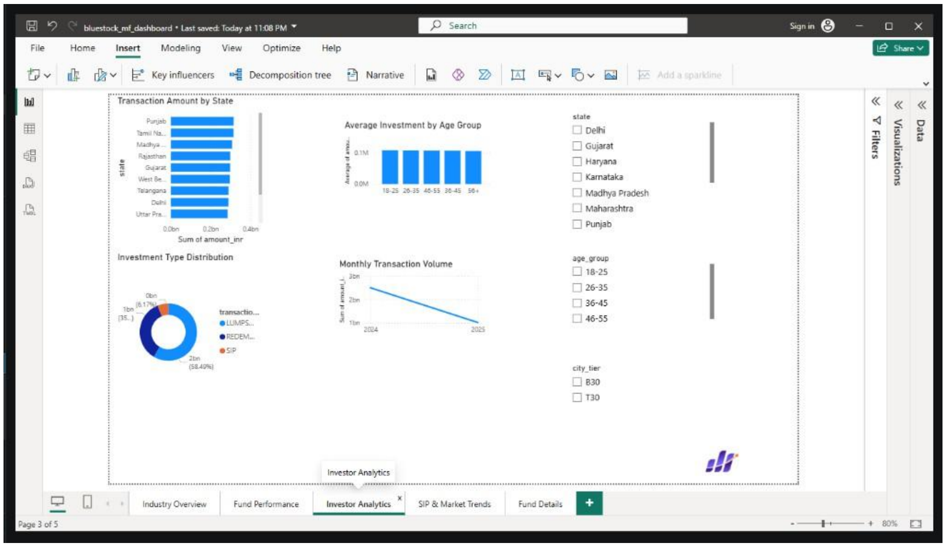
This page analyzes investor participation, demographics, and transaction behavior.

Key Components

- Transaction Amount by State
- Investment Type Distribution
- Age Group Analysis
- Monthly Transaction Volume
- State Filters
- Age Group Filters
- City Tier Filters

Insights

- The 26–35 age group represented the largest investor segment.
- SIP transactions accounted for a significant share of investments.
- Geographic participation varied across states.
- Investor activity showed consistent growth throughout the analysis period.



8.5 Dashboard Page 4: SIP and Market Trends

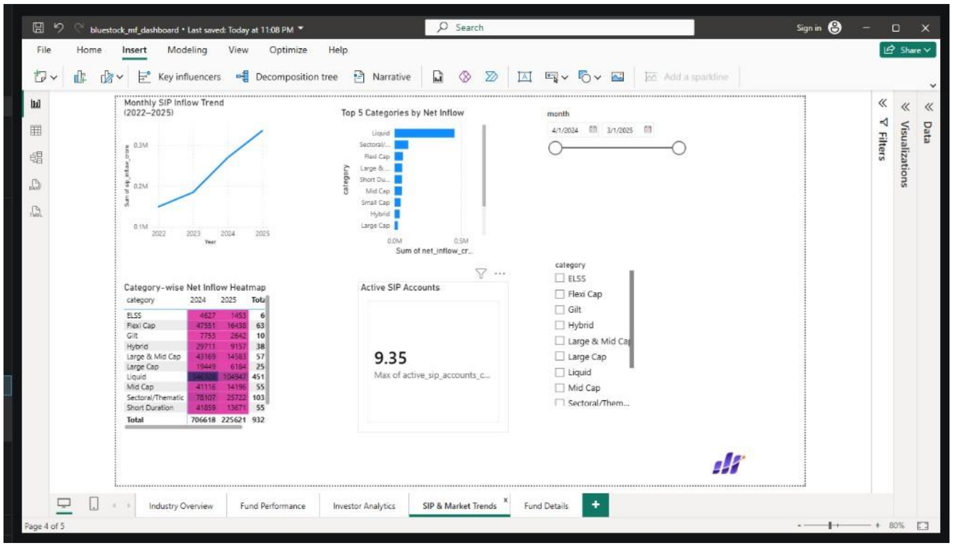
This page focuses on industry-wide investment patterns and market participation trends.

Key Components

- SIP Inflow Trend Analysis
- Category-wise Inflow Heatmap
- Top Categories by Net Inflow
- Active SIP Account Growth
- Month and Category Filters

Insights

- SIP inflows increased steadily over the study period.
- Flexi Cap and Mid Cap categories attracted strong investor interest.
- Active SIP accounts experienced significant growth.
- Category-level trends highlighted changing investor preferences.



8.6 Dashboard Features

The dashboard includes several interactive features:

Slicers

- Fund House
- Category
- Investment Plan
- State
- Age Group
- City Tier
- Month

Interactive Visualizations

- KPI Cards
- Line Charts
- Bar Charts
- Scatter Plots
- Donut Charts
- Heatmaps
- Tables

User Benefits

- Real-time analytical exploration.
- Easy identification of trends and anomalies.
- Improved decision-making through visual insights.
- Simplified access to complex financial metrics.

8.7 Dashboard Outcomes

The Power BI dashboard successfully transformed raw data and analytical outputs into an intuitive business intelligence solution. The dashboard provides stakeholders with a centralized platform to evaluate fund performance, monitor investor behavior, assess market trends, and support investment decision-making through interactive visualizations and data-driven insights.

9. Key Findings

The analysis of mutual fund datasets generated several important insights related to fund performance, investor behavior, market trends, and portfolio risk.

Major Findings

1. The mutual fund industry experienced consistent growth in Assets Under Management (AUM) throughout the analysis period, indicating increasing investor participation.
2. Monthly SIP inflows showed a strong upward trend, reflecting growing investor confidence in systematic investment strategies.
3. The 26–35 age group represented the largest segment of investors, highlighting strong participation from young professionals and retail investors.
4. Moderate-risk mutual funds achieved the highest risk-adjusted performance based on Sharpe Ratio analysis.
5. Several funds generated positive Alpha values, demonstrating their ability to outperform benchmark indices.
6. Value at Risk (VaR) and Conditional Value at Risk (CVaR) analysis identified funds with higher downside risk exposure during adverse market conditions.
7. Rolling Sharpe Ratio analysis revealed that fund performance varies significantly across different market cycles.
8. Investor cohort analysis indicated differences in investment behavior and contribution levels across investor groups.
9. Sector concentration analysis showed that some portfolios were heavily exposed to specific sectors, increasing concentration risk.
10. The fund recommendation system successfully identified suitable funds based on investor risk preferences and risk-adjusted returns.

Overall, the findings demonstrate that combining performance metrics, risk analytics, and investor behavior analysis provides a more comprehensive understanding of mutual fund performance and investment opportunities.

10. Recommendations

Based on the analysis conducted throughout the project, the following recommendations are proposed:

For Investors

1. Consider risk-adjusted performance metrics such as Sharpe Ratio and Sortino Ratio in addition to absolute returns when evaluating mutual funds.
2. Adopt Systematic Investment Plans (SIPs) for long-term wealth creation and disciplined investing.
3. Diversify investments across multiple fund categories and sectors to reduce portfolio risk.
4. Regularly review portfolio performance and rebalance investments according to financial goals and risk tolerance.
5. Avoid concentrating investments in funds with excessive sector exposure.

For Fund Managers

1. Monitor downside risk indicators such as VaR and Maximum Drawdown to improve risk management practices.
2. Focus on generating consistent Alpha through effective portfolio management strategies.
3. Encourage investor retention through targeted engagement with at-risk SIP investors.
4. Improve portfolio diversification to reduce sector concentration risk.

For Financial Institutions

1. Utilize advanced analytics and dashboard solutions for continuous performance monitoring.
2. Implement investor segmentation and cohort analysis to enhance customer engagement strategies.
3. Develop recommendation systems to provide personalized investment guidance.

By incorporating these recommendations, investors and financial institutions can make more informed decisions and improve overall investment outcomes.

11. Limitations

Although the project provides valuable insights into mutual fund performance and investor behavior, certain limitations should be acknowledged.

1. The analysis is based on historical data, which may not accurately predict future market performance.
2. The recommendation system uses a simplified rule-based approach and does not incorporate advanced machine learning techniques.
3. Benchmark comparisons were limited to the available benchmark datasets.
4. Portfolio holdings and sector allocations may change over time, affecting risk analysis results.
5. External economic factors such as inflation, interest rates, and geopolitical events were not explicitly incorporated into the analysis.
6. The project focuses on a selected set of mutual fund schemes and may not represent the entire mutual fund industry.
7. Certain assumptions were made during data cleaning and preprocessing to handle missing or inconsistent records.

Despite these limitations, the project successfully demonstrates the application of financial analytics, risk assessment, and business intelligence techniques in the mutual fund domain.

12. Conclusion

The Mutual Fund Analytics Capstone Project successfully demonstrated the end-to-end application of data analytics, financial modeling, risk assessment, and business intelligence techniques within the mutual fund industry.

The project involved the collection, cleaning, validation, and analysis of multiple datasets related to mutual funds, investor transactions, portfolio holdings, SIP inflows, and benchmark indices. Through Exploratory Data Analysis (EDA), performance evaluation, advanced risk analytics, and dashboard development, meaningful insights were generated regarding fund performance, investor behavior, and market trends.

Key financial metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, VaR, and CVaR were calculated to evaluate fund performance and risk. A fund recommendation system was developed to support investment decision-making based on investor risk profiles. Additionally, an interactive Power BI dashboard was created to provide stakeholders with a comprehensive and user-friendly analytical platform.

The project highlights the importance of combining data-driven insights with financial analysis to support informed investment decisions. It also demonstrates the practical value of integrating Python, SQL, and Power BI within a real-world financial analytics workflow.

Overall, the Mutual Fund Analytics Capstone Project achieved its objectives and provides a strong foundation for future enhancements involving machine learning, predictive analytics, and automated investment advisory solutions.